

Validation of REGFM_B1

The REGFM_B1 blackbox model in PSCAD and REGFM_B1 model in PSSE were tested using the test system depicted in figure 1 under various cases. Table I explains disturbances simulated to validate the model.

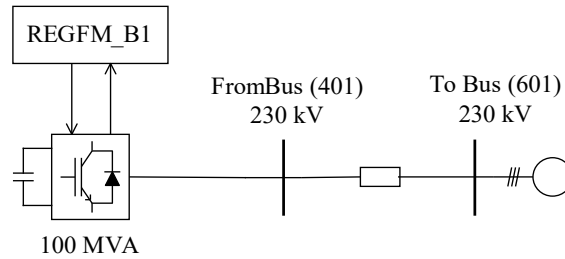


Figure 1 Test system used for model validation.

TABLE 1 Simulated disturbances with Fflag = 1 and ESflag = 1

Disturbances	Scenarios
1. Voltage Step Down	Voltage stepped down to 0.95 pu after 3 seconds
2. Voltage Step Up	Voltage stepped up to 1.05 pu after 3 seconds
3. Frequency Step Down	Frequency stepped down to 59.8 Hz after 3 seconds
4. Frequency Step Up	Frequency stepped up to 60.2 Hz after 3 seconds
5. Short-Term Fault	0.1 second fault at 3 second with 0.189e-3 pu fault
6. Long-Term Fault	0.5 second fault at 3 second with 0.189e-3 pu fault
7. Long-Term High Impedance Fault	0.5 second fault at 3 second with 0.0567 pu fault

Case 1: Voltage Step Down from 1.0 p.u. to 0.95 p.u.

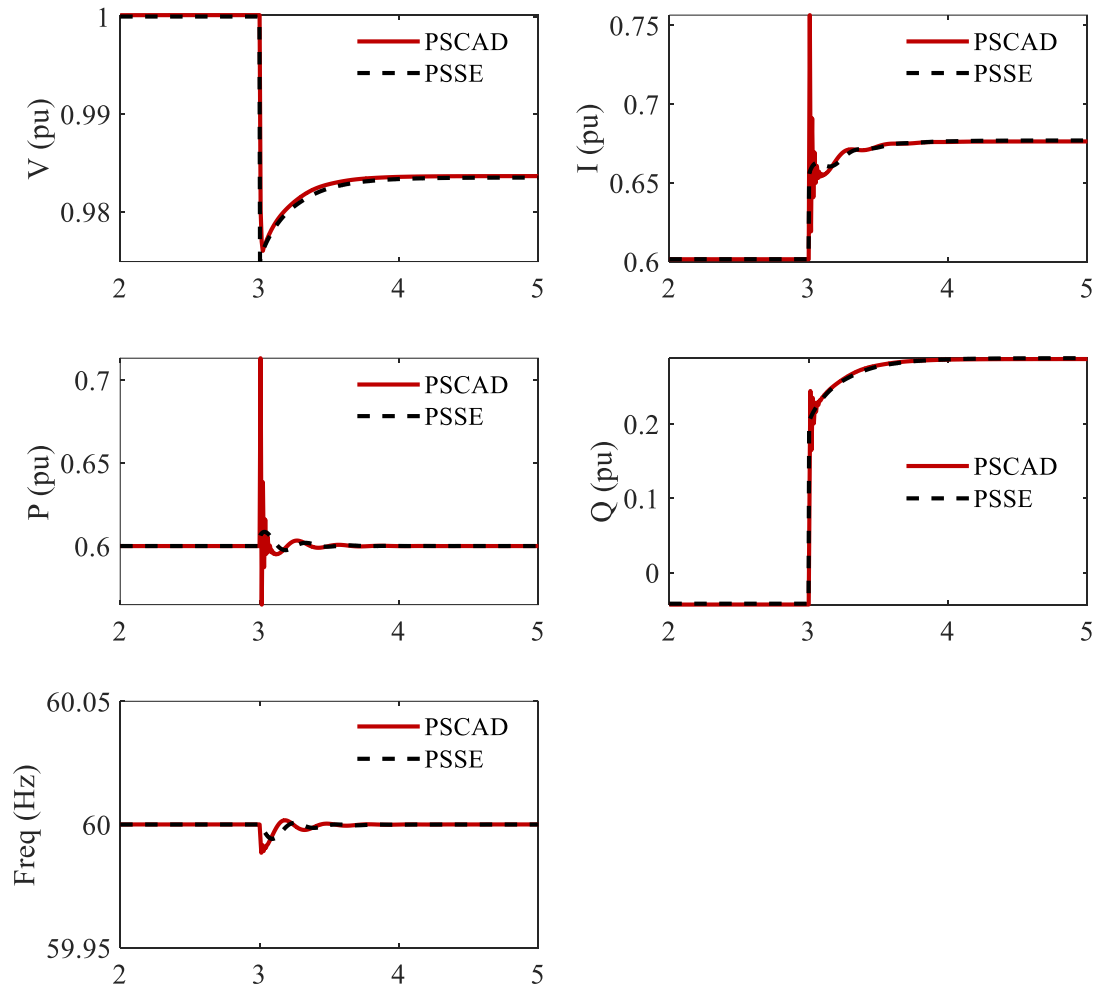


Figure 2: REGFM_B1 output parameters from PSCAD and PSSE simulation results

Case 2: Voltage Step Up from 1.0 p.u. to 1.05 p.u.

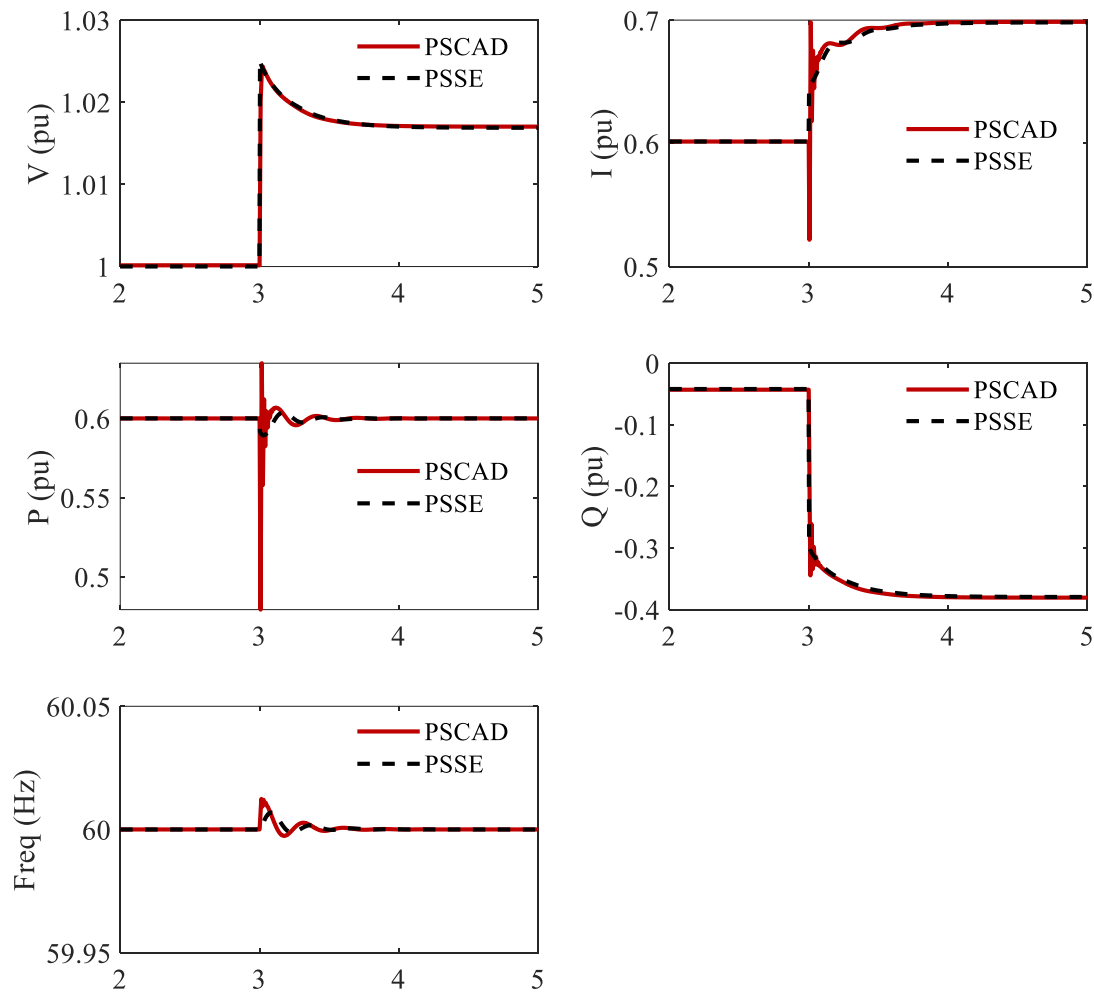


Figure 3: REGFM_B1 output parameters from PSCAD and PSSE simulation results

Case 3: Frequency Step Down from 60 Hz to 59.8 Hz

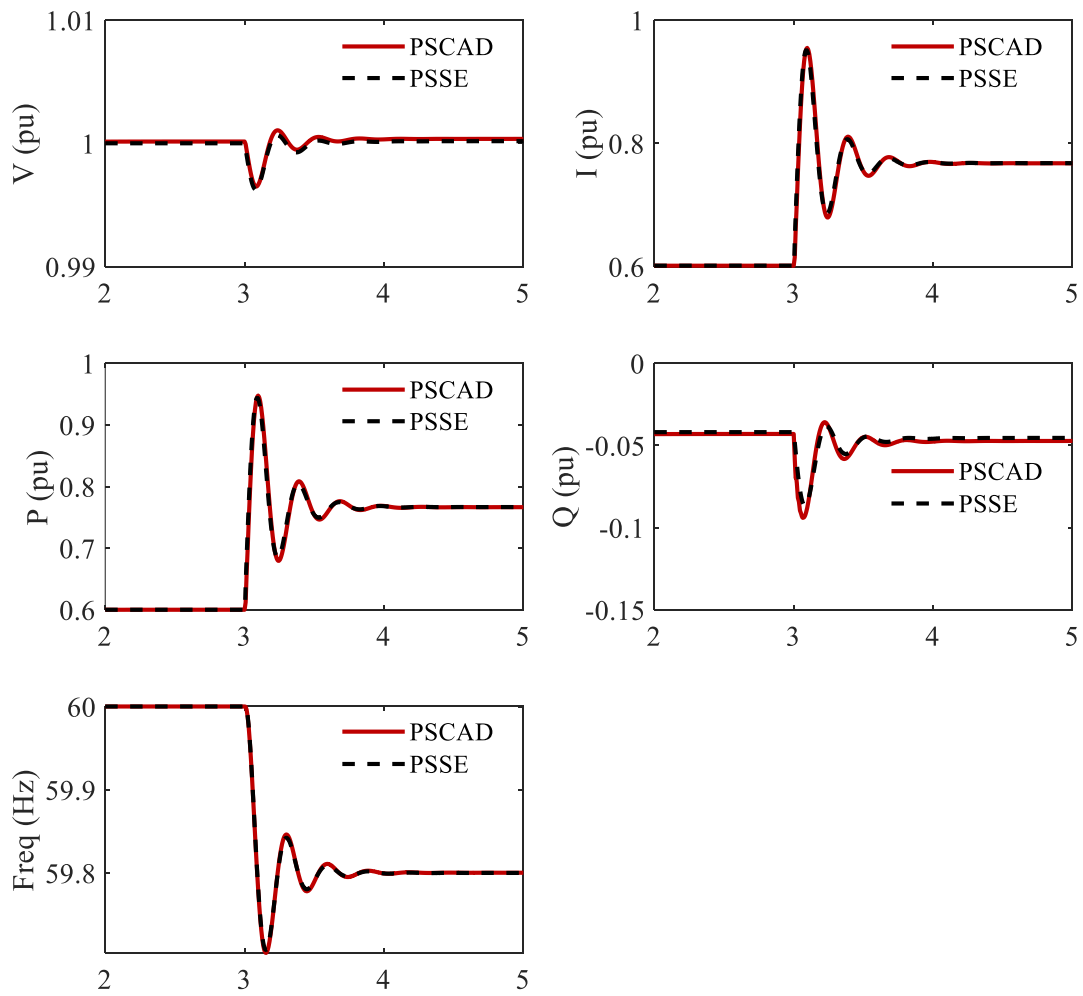


Figure 4: REGFM_B1 output parameters from PSCAD and PSSE simulation results

Case 4: Frequency Step Up from 60 Hz to 60.2 Hz

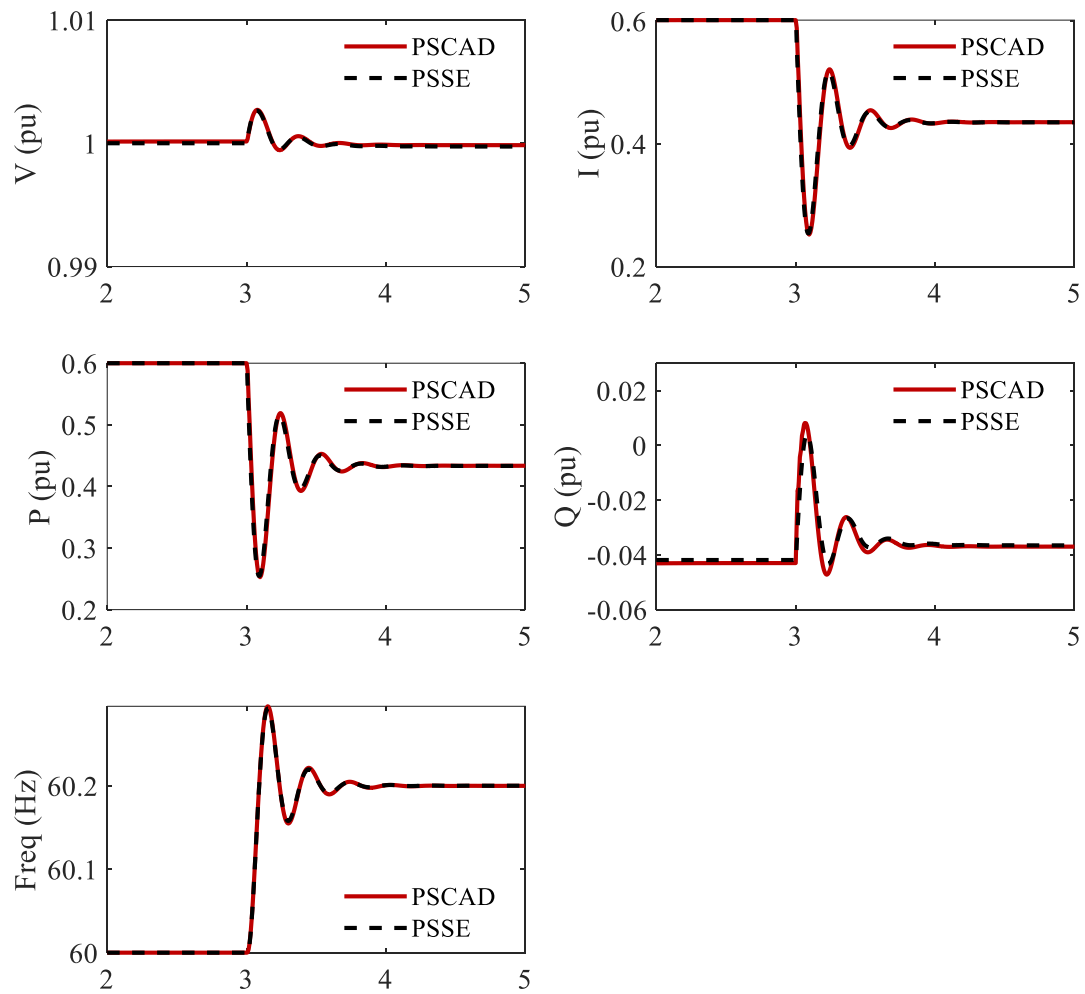


Figure 5: REGFM_B1 output parameters from PSCAD and PSSE simulation results

Case 5: Short-Term Fault (0.1 second)

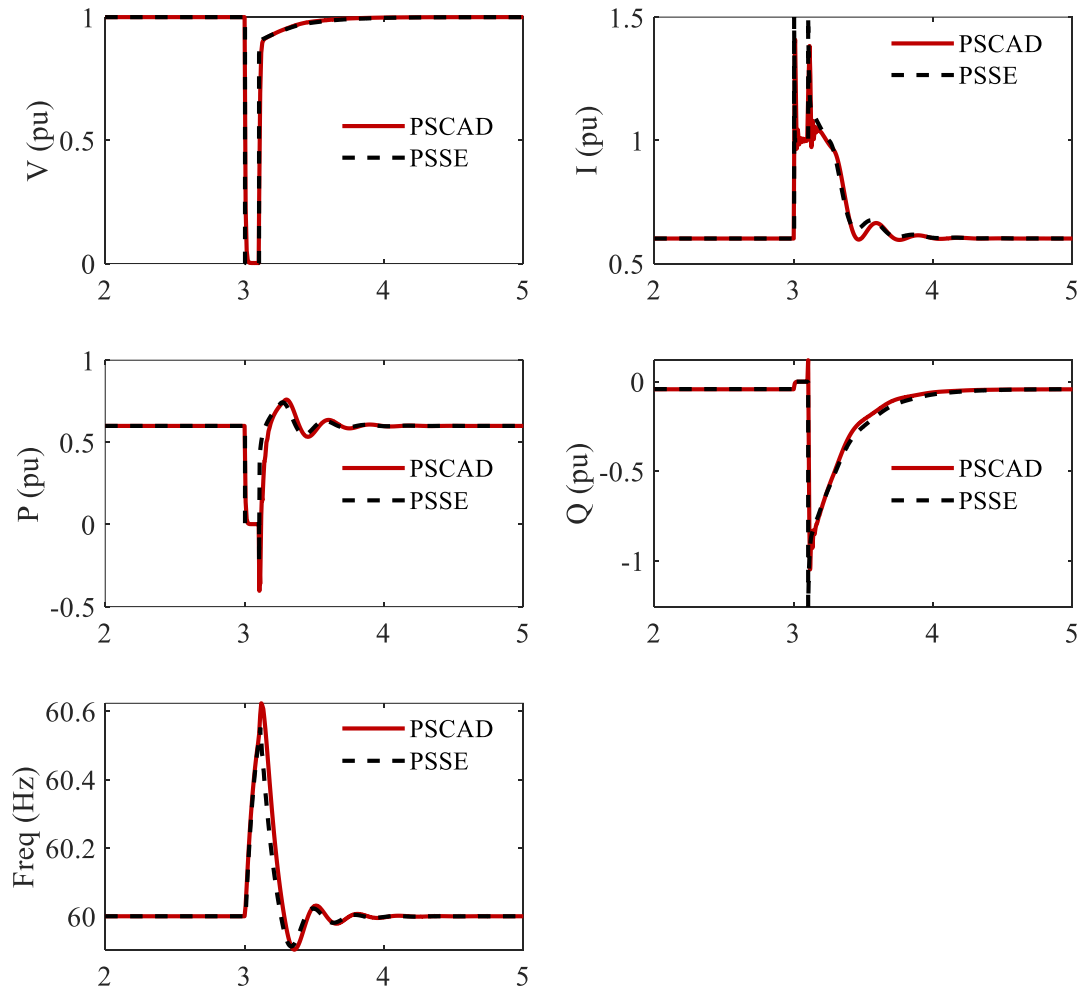


Figure 6: REGFM_B1 output parameters from PSCAD and PSSE simulation results

Case 6: Long-Term Fault (0.5 seconds)

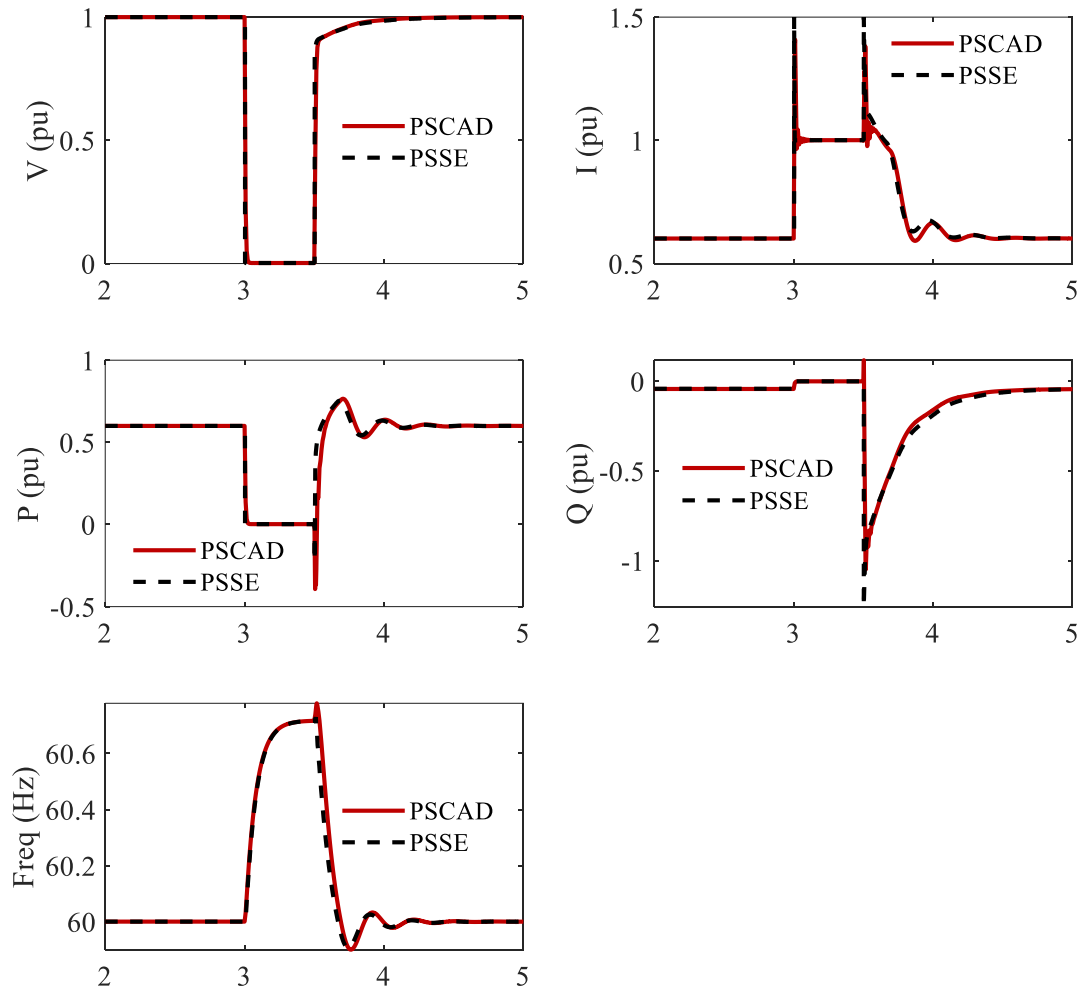


Figure 7: REGFM_B1 output parameters from PSCAD and PSSE simulation results

Case 7: Long-Term High-Impedance Fault (0.5 seconds)

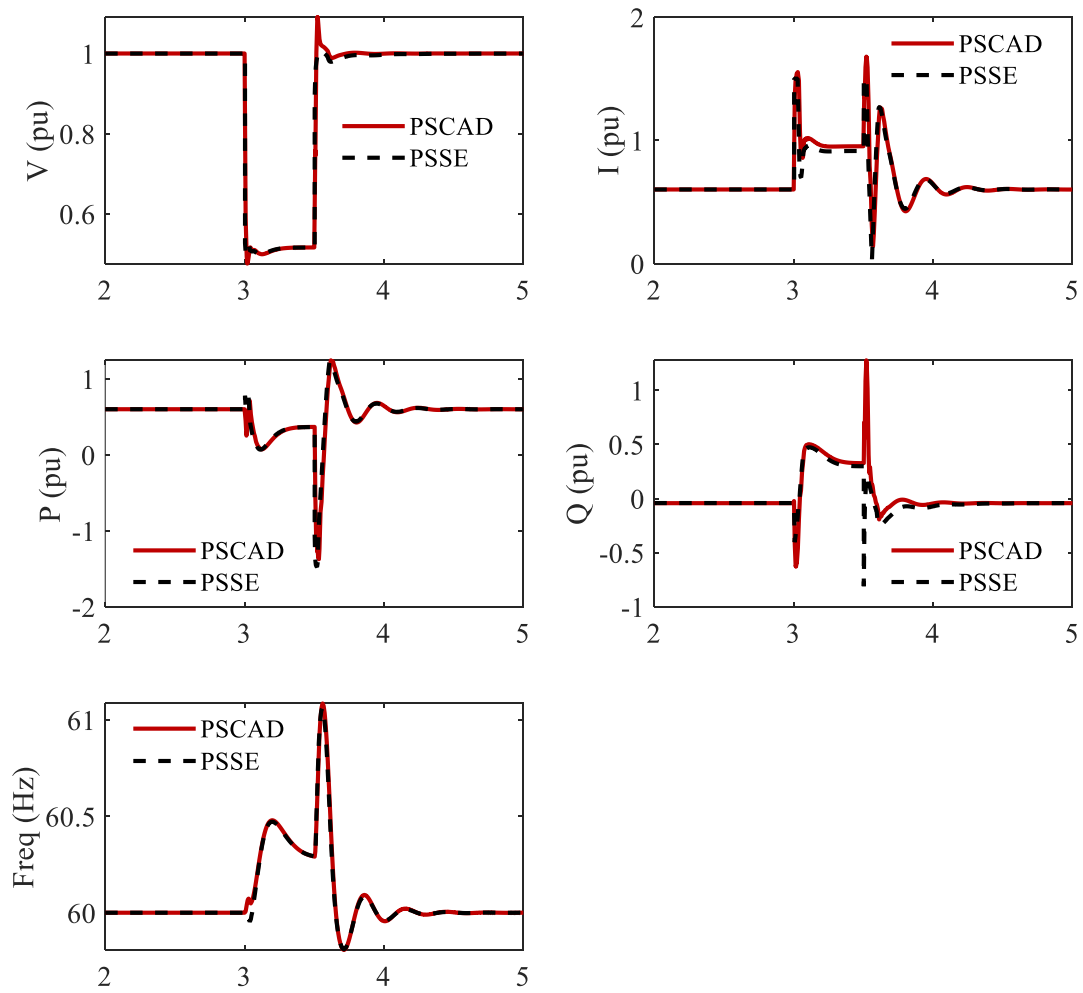


Figure 8: REGFM_B1 output parameters from PSCAD and PSSE simulation results